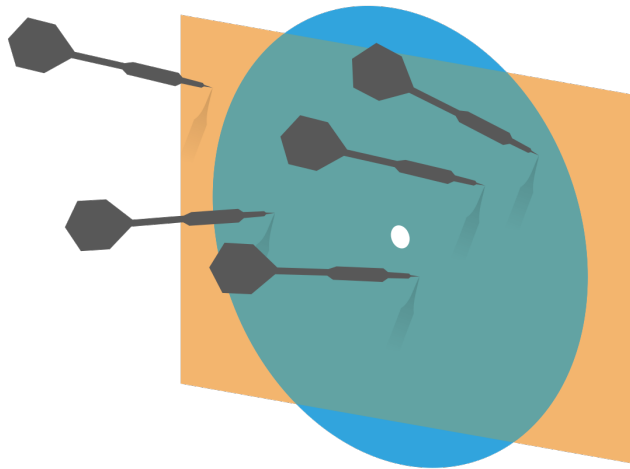


Probability and Statistics for Computer Science



“In statistics we apply probability
to draw conclusions from data.”
---Prof. J. Orloff

Credit: wikipedia

Objectives

✱ Review Sample mean, CI

✱ t-distribution (I)

✱ Bootstrap simulation

✱ Hypothesis test

$$\bar{X} \text{ or } X^{(n)}$$

$$= \frac{\sum X_i}{N}$$

$$\text{var}[\bar{X}] = \left(\frac{1}{N}\right)^2 \text{var}(\sum X_i)$$

$$= \left(\frac{1}{N}\right)^2 \cdot N \cdot \text{var}[X_i]$$

$$= \frac{\text{var}[X_i]}{N}$$

$$[x_0 \pm b \text{ std err}]$$

$$\hookrightarrow \frac{\text{std err}(\{x\})}{\sqrt{N}}$$

$$\text{std}[\bar{X}] = \frac{1}{\sqrt{N}} \cdot \boxed{\text{std}[X_i]}$$

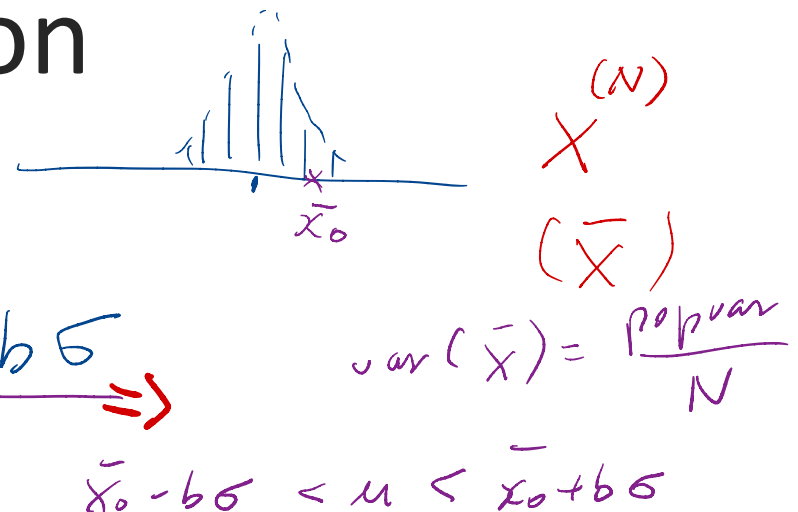
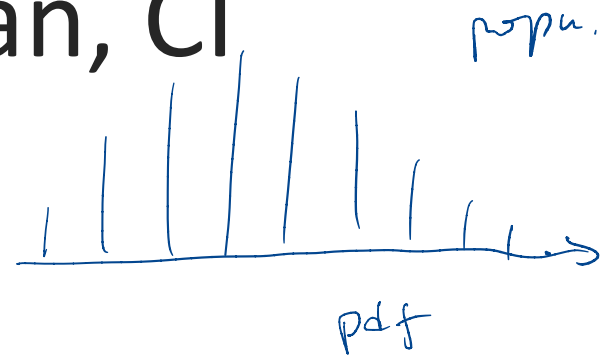
Objectives

✱ Review Sample mean, CI

✱ t-distribution (I)

✱ Bootstrap simulation

✱ Hypothesis test



$$\mu - b\sigma < \bar{x}_0 < \mu + b\sigma \Rightarrow$$

$$\Downarrow$$

$$-b < \hat{\bar{x}}_0 < b$$

$$\bar{x}_0 - b\sigma < \mu < \bar{x}_0 + b\sigma$$

Review with Questions

- 1) Why is sample mean $X^{(N)}$ a random variable?
- 2) Is $E[X^{(N)}] = \text{mean}(\{x\})$? $\{x\}$ is some realized data of size N , drawn from the population $\{X\}$ with replacement.
- A True
- B False
- 3) What is the distribution of $X^{(N)}$? if $N \rightarrow \infty$
- 4) What are $E[X^{(N)}]$, $\text{var}[X^{(N)}]$?

About the distribution of $X^{(N)}$

if $N \rightarrow \infty$ $X^{(N)} \sim \text{Normal}(\mu, \sigma)$ ^{mean $\{x\}$}

$$\mu = E[X^{(N)}] = \text{popmean}$$

$$\sigma = \text{std}[X^{(N)}] = \frac{\text{popstd}}{\sqrt{N}}$$

if $X^{(1)}$ is from a Normal like population,

$$T = \frac{\text{mean}\{x\} - \text{popmean}}{\text{stderr}\{x\}} \sim t \text{ distribution with DOF } N-1$$

$X^{(N)}$

$$\text{stderr} = \frac{\text{stdunk}}{\sqrt{N}}$$

i.i.d. $X^{(1)}$

Sample mean of a population

- ✱ The sample mean is the average of **IID** samples

$$\underline{X^{(N)}} = \frac{1}{N}(X_1 + X_2 + \dots + X_N) = \text{mean}(\{x\})$$

- ✱ By linearity of the expectation and the fact the sample items are identically drawn from the same population with replacement

$$E[X^{(N)}] = \frac{1}{N}(E[X^{(1)}] + E[X^{(1)}].. + E[X^{(1)}]) = E[X^{(1)}]$$

Expected value of one random sample is the population mean

- ✱ Since each sample is drawn uniformly from the population

$$\sum x \cdot p(x) = \sum x \cdot \frac{1}{N_p} = \frac{1}{N_p} \cdot \sum x$$

$$E[X^{(1)}] = \text{popmean}(\{X\})$$

therefore $E[X^{(N)}] = \text{popmean}(\{X\})$

const

- ✱ We say that $X^{(N)}$ is an unbiased estimator of the population mean.

$$E[R_E] = \text{Target Val}$$

$\uparrow$
 R_E is random variable

const

Standard deviation of the sample mean

- ✱ We can also rewrite another result from the lecture on the weak law of large numbers

$$\text{var}[X^{(N)}] = \frac{\text{popvar}(\{X\})}{N}$$

Handwritten note: $\text{std}[X^{(N)}] = \sqrt{\text{var}[X^{(N)}]}$

- ✱ The standard deviation of the sample mean

$$\text{std}[X^{(N)}] = \frac{\text{popstd}(\{X\})}{\sqrt{N}}$$

- ✱ But we need the population standard deviation in order to calculate the $\text{std}[X^{(N)}]$!

Adjusted estimate of population standard deviation & the Stderr

✱ We use the following to estimate $popsd(\{X\})$

$$v = stdunbiased(\{x\}) = \sqrt{\frac{1}{N-1} \sum_{x_i \in \text{sample}} (x_i - \text{mean}(\{x_i\}))^2}$$

$\therefore E[v^2] = \text{var}[\{X\}]$

✱ Then the **standard error** is an estimate of

$$std[X^{(N)}]$$

recall : $std[X^{(N)}] = \frac{popsd(\{X\})}{\sqrt{N}}$

approximated by

$$\frac{popsd(\{X\})}{\sqrt{N}} \doteq \frac{stdunbiased(\{x\})}{\sqrt{N}} = \boxed{stderr(\{x\})}$$

Standard error: election poll

	DATES	POLLSTER	SAMPLE	RESULT	NET RESULT
U.S. Senate Miss.	NOV 25, 2018	C+ Change Research	1,211 LV	Espy 46% 51% Hyde-Smith	Hyde-Smith +5

✱ What is the estimate of the percentage of votes for Hyde-Smith? 51%

$X_i = \begin{cases} 1 & \text{for Smith} \\ 0 & \text{otherwise} \end{cases}$

Number of sampled voters who selected Ms. Smith is:

$$1211(0.51) \approx 618$$

$$\sum X_i = \text{total votes for Smith}$$

Number of sampled voters who didn't select Ms. Smith was

$$1211(0.49) \approx 593$$

$$\bar{X}^{(N)} = \frac{\sum_{i=1}^N X_i}{N} \quad N = 1211$$

Standard error: election poll

✱ $stdunbiased(\{x\})$ { 1, 0, 1, 1, - - - }

$$= \sqrt{\frac{1}{1211 - 1} (618(1 - 0.51)^2 + 593(0 - 0.51)^2)} = 0.5001001$$

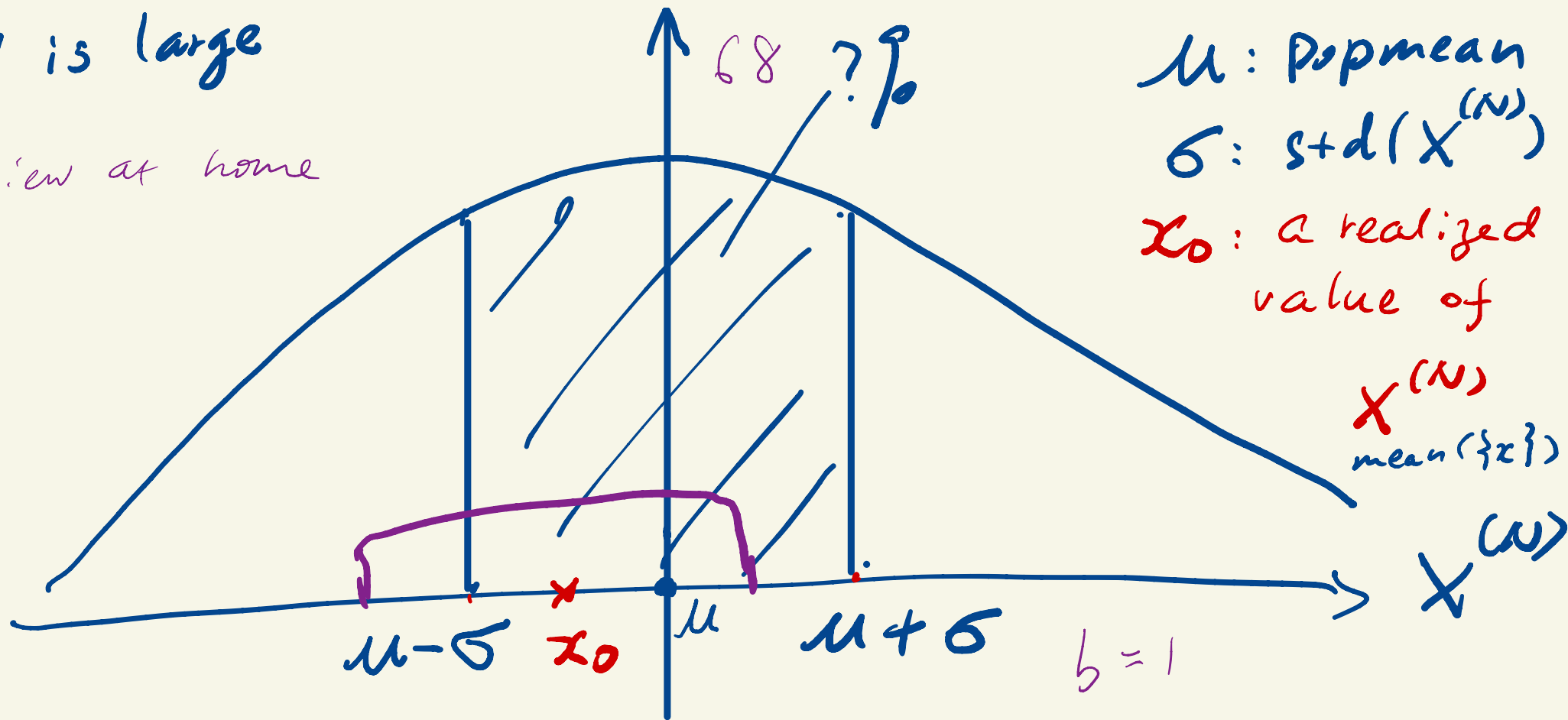
✱ $stderr(\{x\})$

$$= \frac{0.5001001}{\sqrt{1211}} \simeq 0.0144$$

$N = 1211$

N is large

Review at home



How much chance is it
that x_0 is

located in $[\mu - \sigma, \mu + \sigma]$?

If $\mu - \sigma < x_0 < \mu + \sigma$
then $\Rightarrow$

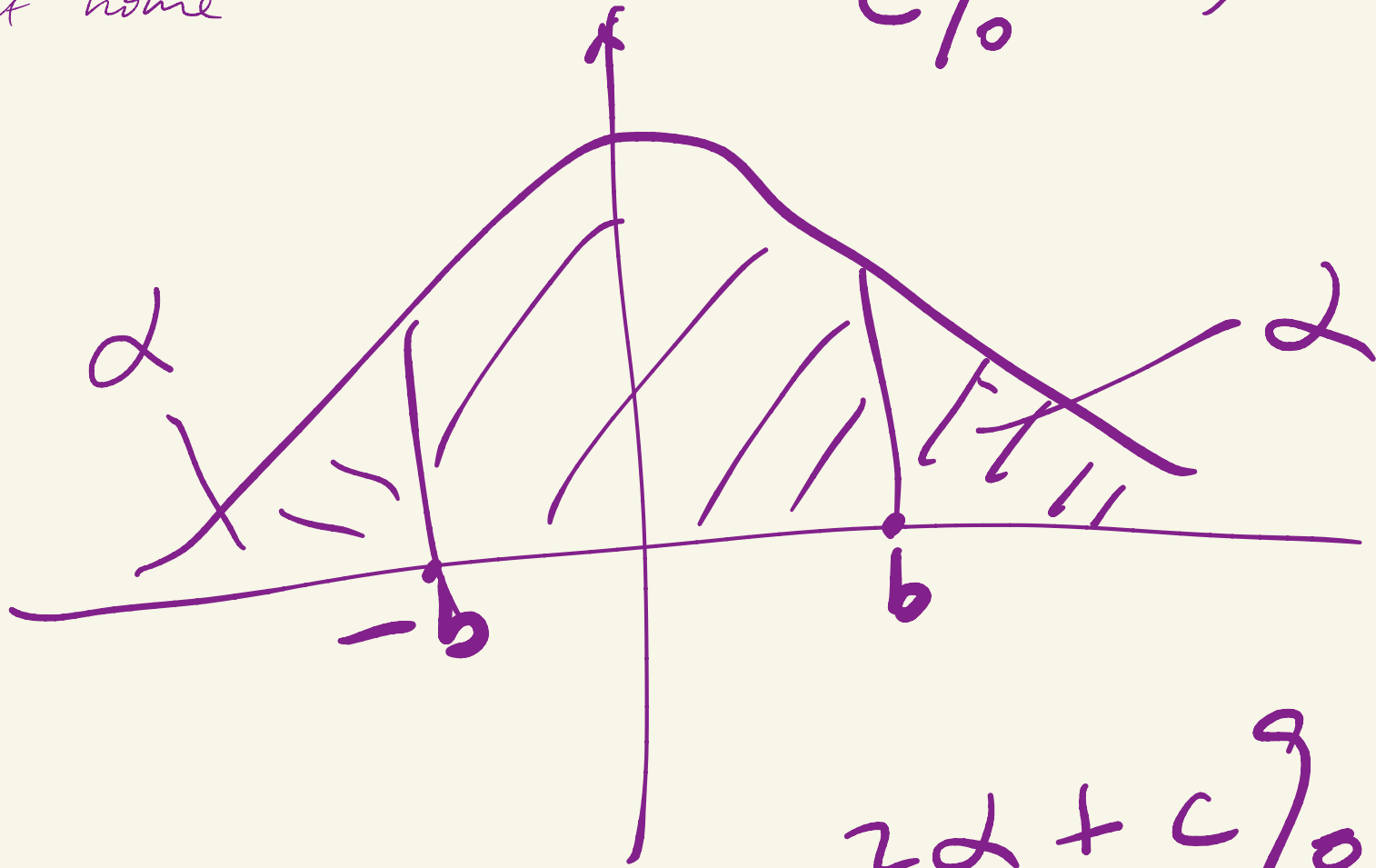
$$x_0 - \sigma < \mu < x_0 + \sigma$$

$\sigma \sim \text{stderr}$

Review at home

c%

$N(0, 1)$

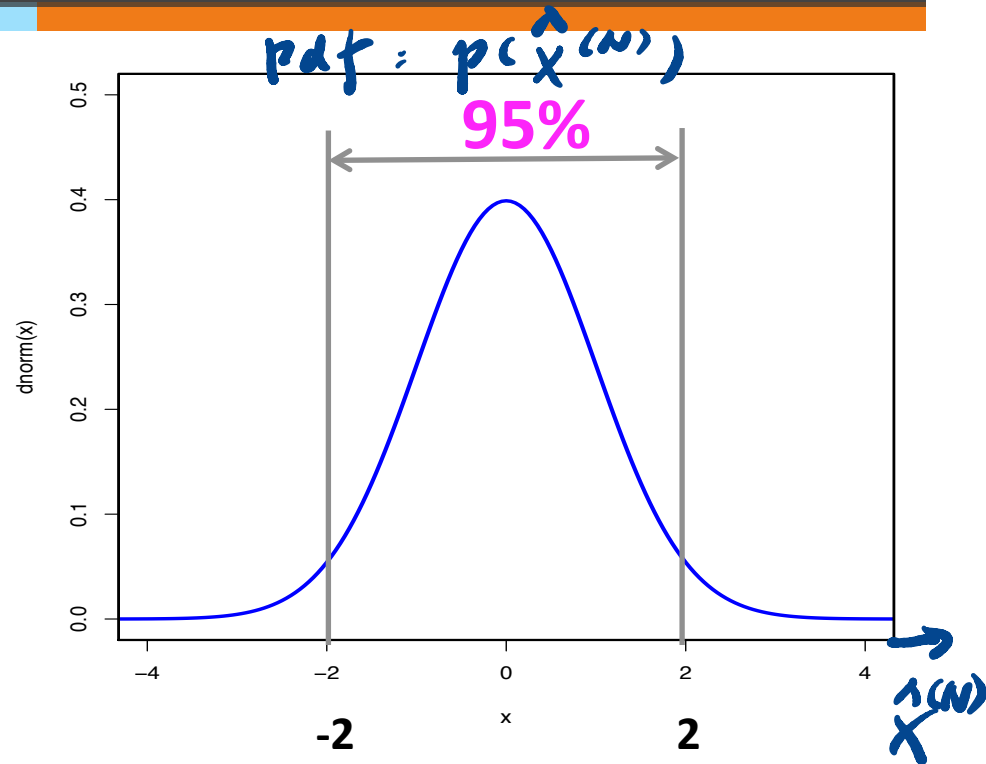


$$\bar{X}_0 \pm b \cdot \text{stderr}$$

$$\underline{\text{CDF}(b) = 1 - \alpha}$$

Confidence intervals

- ✱ Confidence interval for a population mean is defined by fraction
- ✱ Given a percentage, find how many units of strerr it covers.

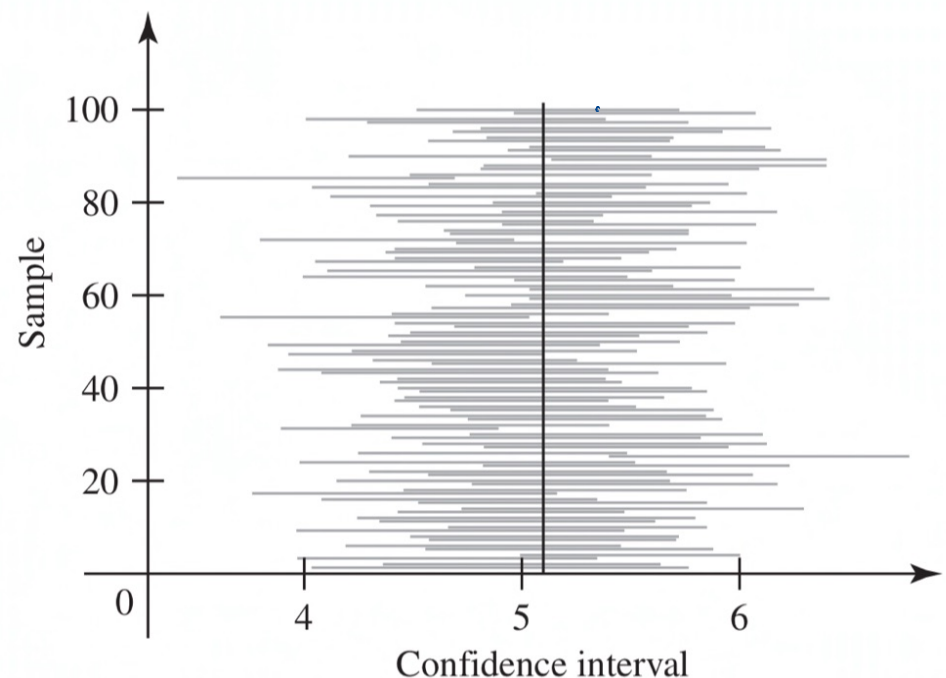


For **95%** of the **realized sample means**,
 the population mean lies in
 [sample mean-2 stderr, sample mean+2 stderr]

Interpreting the confidence intervals

$$\text{mean}(\{x\}) \pm 2 \cdot \text{stdem}$$

Figure 8.5 A sample of one hundred observed 95% confidence intervals based on samples of size 26 from the normal distribution with mean $\mu = 5.1$ and standard deviation $\sigma = 1.6$. In this figure, 94% of the intervals contain the value of μ .



Deyoung Pg 487

$$std(\{x\}) = \sqrt{\frac{1}{N} \underbrace{(\sum (x_i - \bar{x})^2)}_C}$$

Review at home

$$std_{unb}(\{x\}) = \sqrt{\frac{1}{N-1} \underbrace{(\sum (x_i - \bar{x})^2)}_C}$$

$$std_{unb} = \frac{std \cdot \sqrt{N}}{\sqrt{N-1}}$$

$$std_{err} = \frac{std_{unb}}{\sqrt{N}} = \frac{std(\{x\})}{\sqrt{N-1}}$$

Interpreting the standard error

- ✱ **Sample mean** is a random variable and has its own probability distribution, `stderr` is an estimate of sample mean's standard deviation
- ✱ When ***N*** is very large, according to the **Central Limit Theorem**, sample mean is approaching a normal distribution with

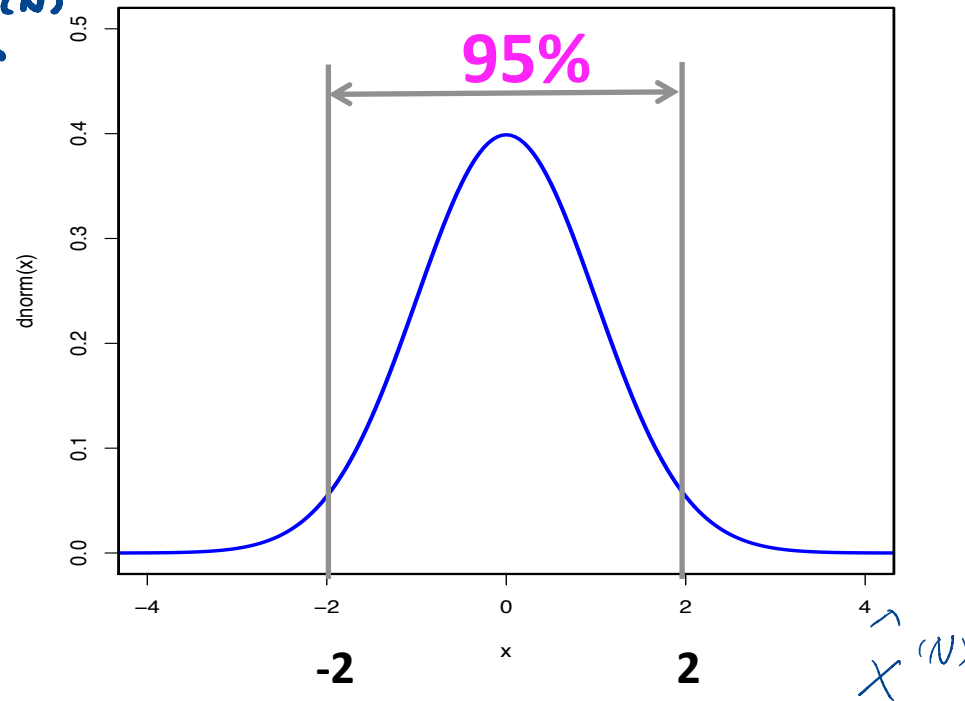
$$X^{(N)} \sim N(\mu, \sigma)$$

$$\mu = \underset{\approx \text{mean}(\{x\})}{\text{popmean}(\{X\})} ; \sigma = \frac{\text{popstd}(\{X\})}{\sqrt{N}} \approx \text{stderr}(\{x\})$$

$$\text{stderr}(\{x\}) = \frac{\text{stdunbiased}(\{x\})}{\sqrt{N}}$$

Confidence intervals

- ✱ Confidence interval ^{part of $\bar{X}^{(N)}$} for a population mean is defined by fraction
- ✱ Given a percentage, find how many units of stderr it covers.



For **95%** of the **realized sample means**,
 the population mean lies in
 [sample mean-2 stderr, sample mean+2 stderr]

Confidence intervals when N is large

- ✱ For about 68% of realized sample means

$$\text{mean}(\{x\}) - \text{stderr}(\{x\}) \leq \text{popmean}(\{X\}) \leq \text{mean}(\{x\}) + \text{stderr}(\{x\})$$

- ✱ For about 95% of realized sample means

$$\text{mean}(\{x\}) - 2\text{stderr}(\{x\}) \leq \text{popmean}(\{X\}) \leq \text{mean}(\{x\}) + 2\text{stderr}(\{x\})$$

- ✱ For about 99.7% of realized sample means

$$\text{mean}(\{x\}) - 3\text{stderr}(\{x\}) \leq \text{popmean}(\{X\}) \leq \text{mean}(\{x\}) + 3\text{stderr}(\{x\})$$

Q. Confidence intervals

✱ What is the 68% confidence interval for a population mean? *And why?*

- A. [sample mean-2stderr, sample mean+2stderr]
- B. [sample mean-stderr, sample mean+stderr]
- C. [sample mean-std, sample mean+std]
of {x} of {x}

Standard error: election poll

	DATES	POLLSTER	SAMPLE	RESULT	NET RESULT
U.S. Senate Miss.	NOV 25, 2018	C+ Change Research	1,211 LV	Espy 46% 51% Hyde-Smith	Hyde-Smith +5

51%

✱ We estimate the population mean as 51% with stderr 1.44%

✱ The 95% confidence interval is
[51%-2×1.44%, 51%+2×1.44%]= [48.12%, 53.88%]

Q.

✱ A store staff mixed their fuji  and gala  apples and they were individually wrapped, so they are indistinguishable. if I pick 30 apples and found 21 fuji , what is my 95% confidence interval to estimate the popmean is 70% for fuji? (hint: $\text{strerr} > 0.05$)

A. $[0.7-0.17, 0.7+0.17]$

B. $[0.7-0.056, 0.7+0.056]$

What if N is small? When is N large enough?

- ✱ If samples are taken from normal distributed population, the following variable is a random variable whose distribution is Student's **t**-distribution with **N-1** degree of freedom.

$$T = \frac{\text{mean}(\{x\}) - \text{popmean}(\{X\})}{\text{stderr}(\{x\})}$$

Handwritten notes:
 $X^{(N)}$ (above $\{x\}$)
 $E[X^{(N)}] = \text{popmean}$ (above $\{X\}$)
 $\text{stderr}(\{x\}) \approx \text{std}(X^{(N)})$ (below denominator)

Degree of freedom is **N-1** due

to this constraint: $\sum_i (x_i - \text{mean}(\{x\})) = 0$

Handwritten notes:
Degroot
Pg. 480

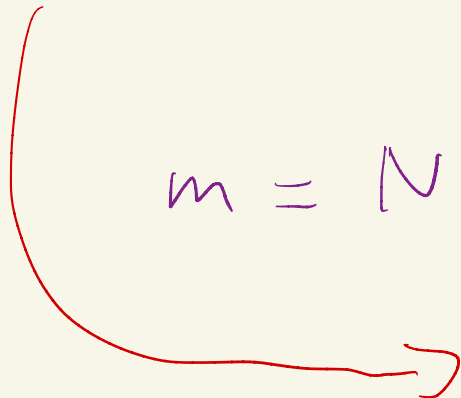
PDF of t-distrib. of m degree

$$\frac{\Gamma\left(\frac{m+1}{2}\right)}{(m\pi)^{1/2} \Gamma\left(\frac{m}{2}\right)} \left(1 + \frac{x^2}{m}\right)^{-(m+1)/2} \quad -\infty < x < \infty.$$

$m = N-1$

mean doesn't exist for $m \leq 1$

$$m = N-1 \leq 1$$


$$N \leq 2$$

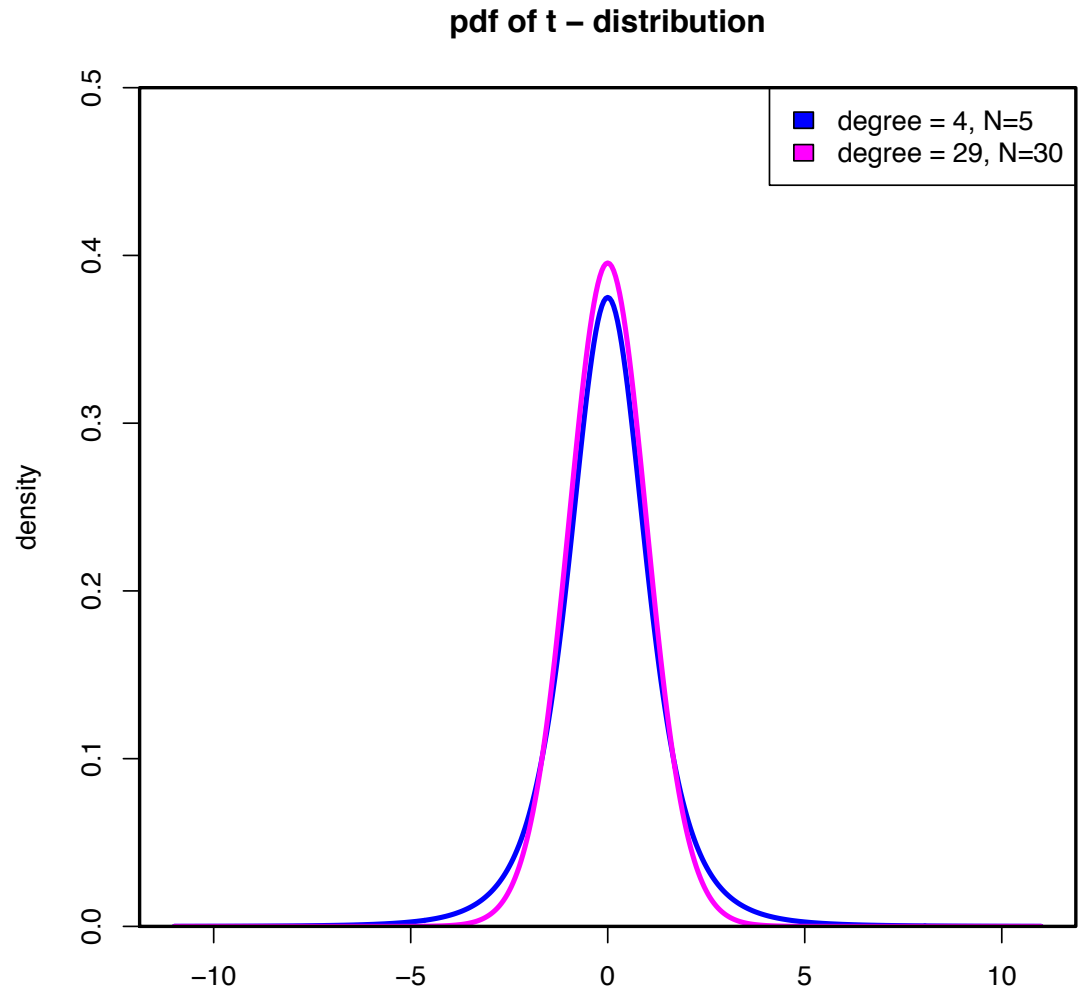
t-distribution is a family of distri. with different degrees of freedom

t-distribution with $N=5$
and $N=30$



Credit :
wikipedia

William Sealy Gosset 1876-1937

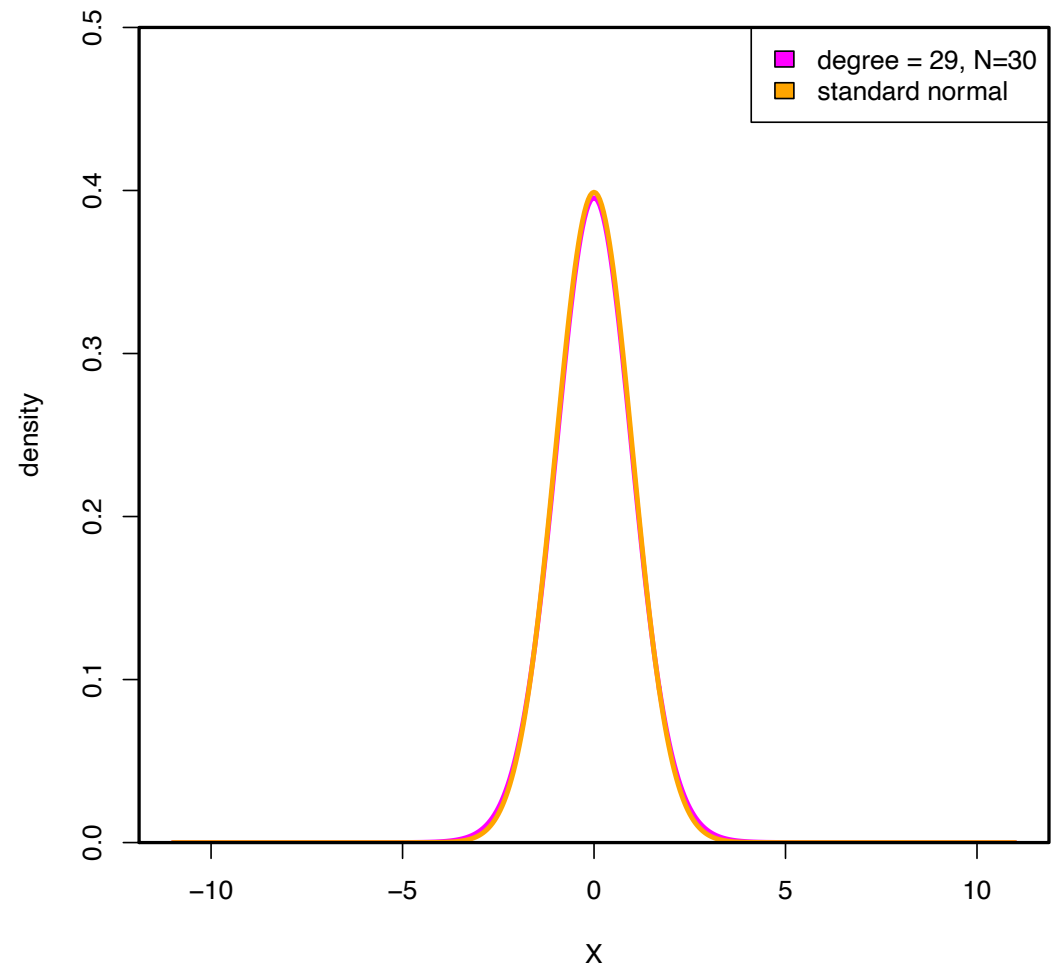


When $N=30$, t-distribution is almost Normal

t-distribution looks very similar to standard normal when $N=30$.

So, $N=30$ is a rule of thumb to decide N is large or not

pdf of t (n=30) and normal distribution



Confidence intervals when $N < 30$

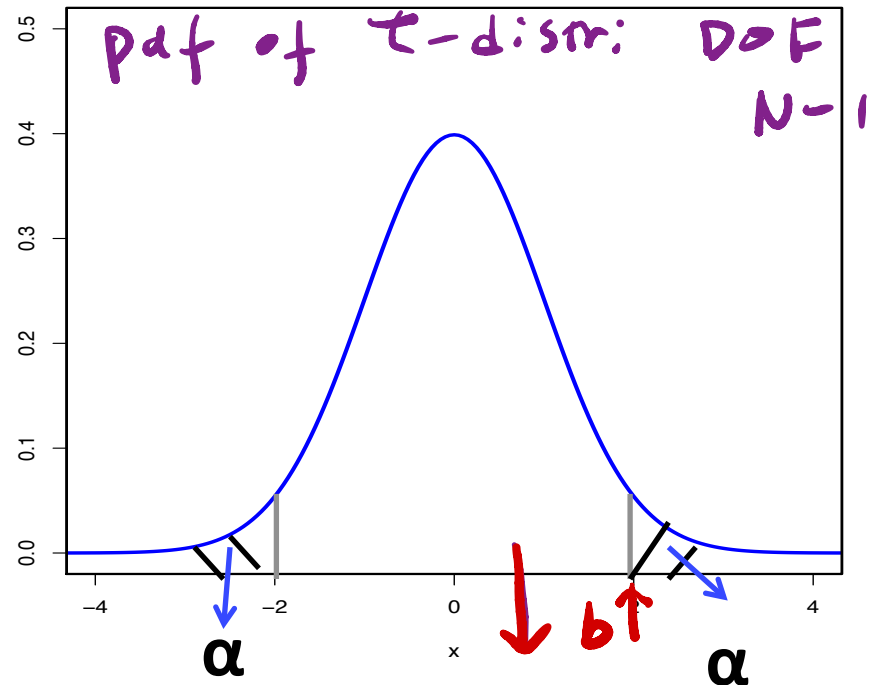
- ✱ If the sample size $N < 30$, we should use t-distribution with its parameter (**the degrees of freedom**) set to $N-1$

t-distr: is also symmetric.

Centered Confidence intervals

- ✱ Centered Confidence interval for a population mean by α value, where

$$P(T \geq b) = \alpha$$



For $1-2\alpha$ of the realized sample means,
the population mean lies in
[sample mean- $b \times \text{stderr}$, sample mean+ $b \times \text{stderr}$]

2 α Confidence Interval

$$P\left(\frac{\text{mean}(\{x\}) - \text{popmean}}{\text{stderr}(\{x\})} \geq b\right)$$

$$= P(\text{popmean} \leq \text{mean}(\{x\}) - b \cdot \text{stderr}(\{x\})) = \alpha$$

$$P\left(\frac{\text{mean}(\{x\}) - \text{popmean}}{\text{stderr}(\{x\})} \leq -b\right)$$

$$= P(\text{popmean} \geq \text{mean}(\{x\}) + b \cdot \text{stderr}(\{x\})) = \alpha$$

$$\alpha = 5\% \rightarrow 1 - 2\alpha = 90\%$$

Q.

✱ The 95% confidence interval for a population mean is equivalent to what $1-2\alpha$ interval?

A. $\alpha = 0.05$

B. $\alpha = 0.025$

C. $\alpha = 0.1$

Sample statistic

- ✱ A **statistic** is a function of a dataset
 - ✱ For example, the mean or median of a dataset is a statistic
- ✱ **Sample statistic**
 - ✱ Is a statistic of the data set that is formed by the realized sample
 - ✱ For example, the realized sample mean

Q. Is this a sample statistic?

✱ The largest integer that is smaller than or equal to the mean of a sample

A. Yes

B. No.

Q. Is this a sample statistic?

✱ The interquartile range of a sample

☒ A. Yes

B. No.

Confidence intervals for other sample statistics

- ✱ **Sample statistic** such as *median* and others are also interesting for drawing conclusion about the population
- ✱ It's often difficult to derive the analytical expression in terms of stderr for the corresponding random variable
- ✱ So, we can use simulation...

A tale of two statisticians

$$\{X\} = \{1, 2, 3, \dots, 12\} \quad N_p = 12$$

The task: use only a subset of $\{X\}$:
 $\{x\}$ with $N = 5$ to estimate the
 $\text{popmean}(\{X\})$ with some confidence
report.

A tale of two statisticians

$$\{X\} = \{1, 2, 3, \dots, 12\} \quad N_p = 12$$

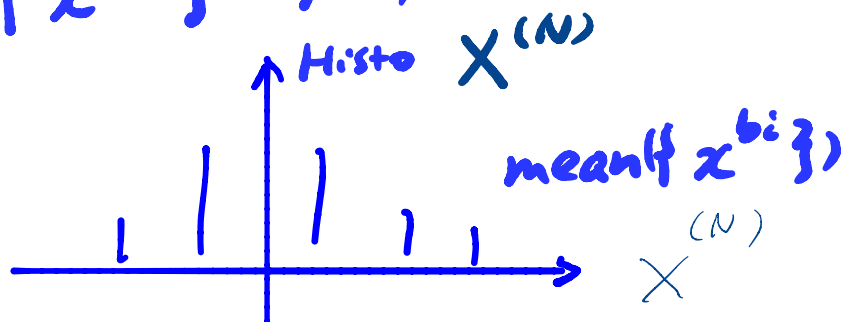
✓ real sample

$$\{x^{b_1}\} = \{1, 4, 5, 7, 11\}$$

$$\{x^{b_2}\} = \{1, 1, 4, 5, 7\} \quad x^{b_1^{(n)}} = \frac{18}{5}$$

$$\{x^{b_3}\} = \{4, 5, 7, 7, 1\}$$

$$\{x^{b_4}\} = \{5, 5, 5, 5, 5\}$$



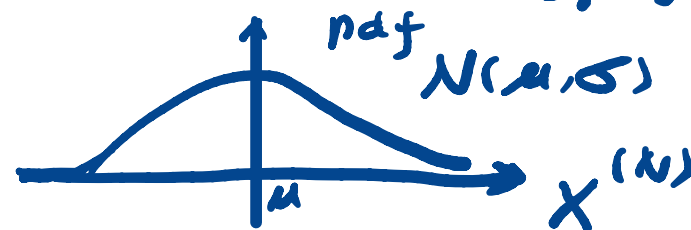
$$\{x\} = \{1, 4, 5, 7, 11\} \quad \text{i.i.d. } x^{(i)} \quad N=5$$

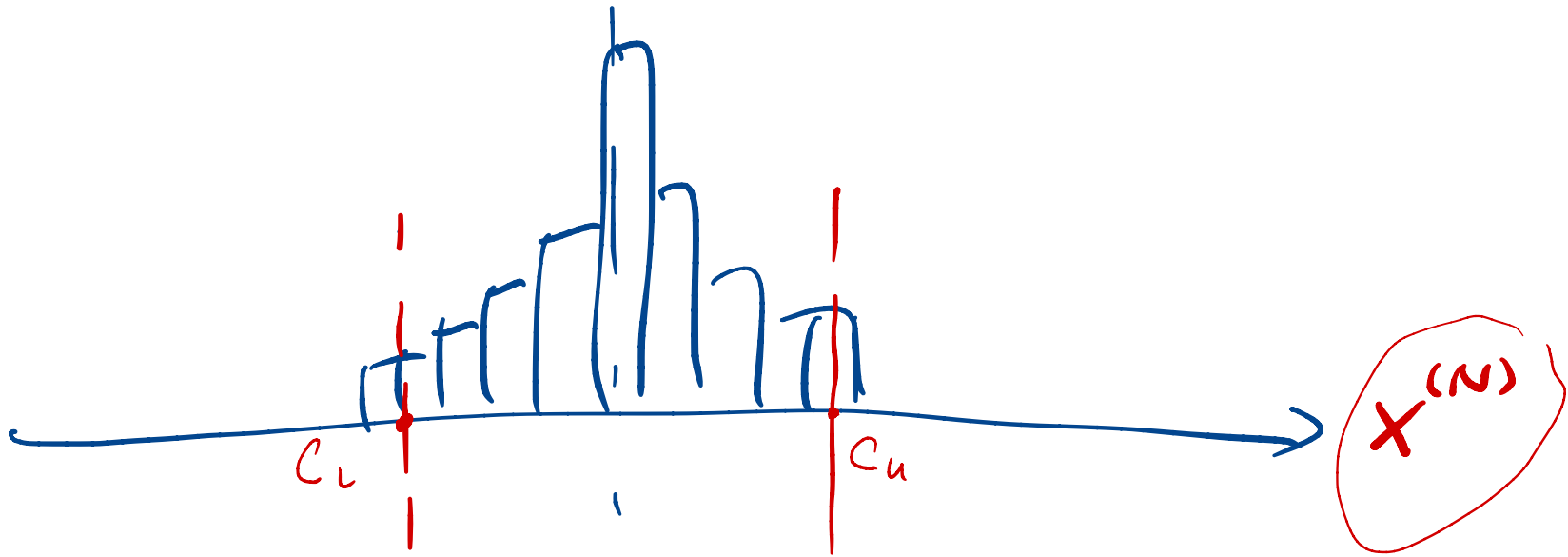
if $N \rightarrow \infty$

$$X^{(N)} \sim N(\mu, \sigma)$$

$$\mu = E[X^{(N)}] = \text{mean}(\{x\})$$

$$\sigma = \text{std}[X^{(N)}] = \text{stderr}(\{x\})$$





$$C_L = \text{Quantile}(2.5\%)$$

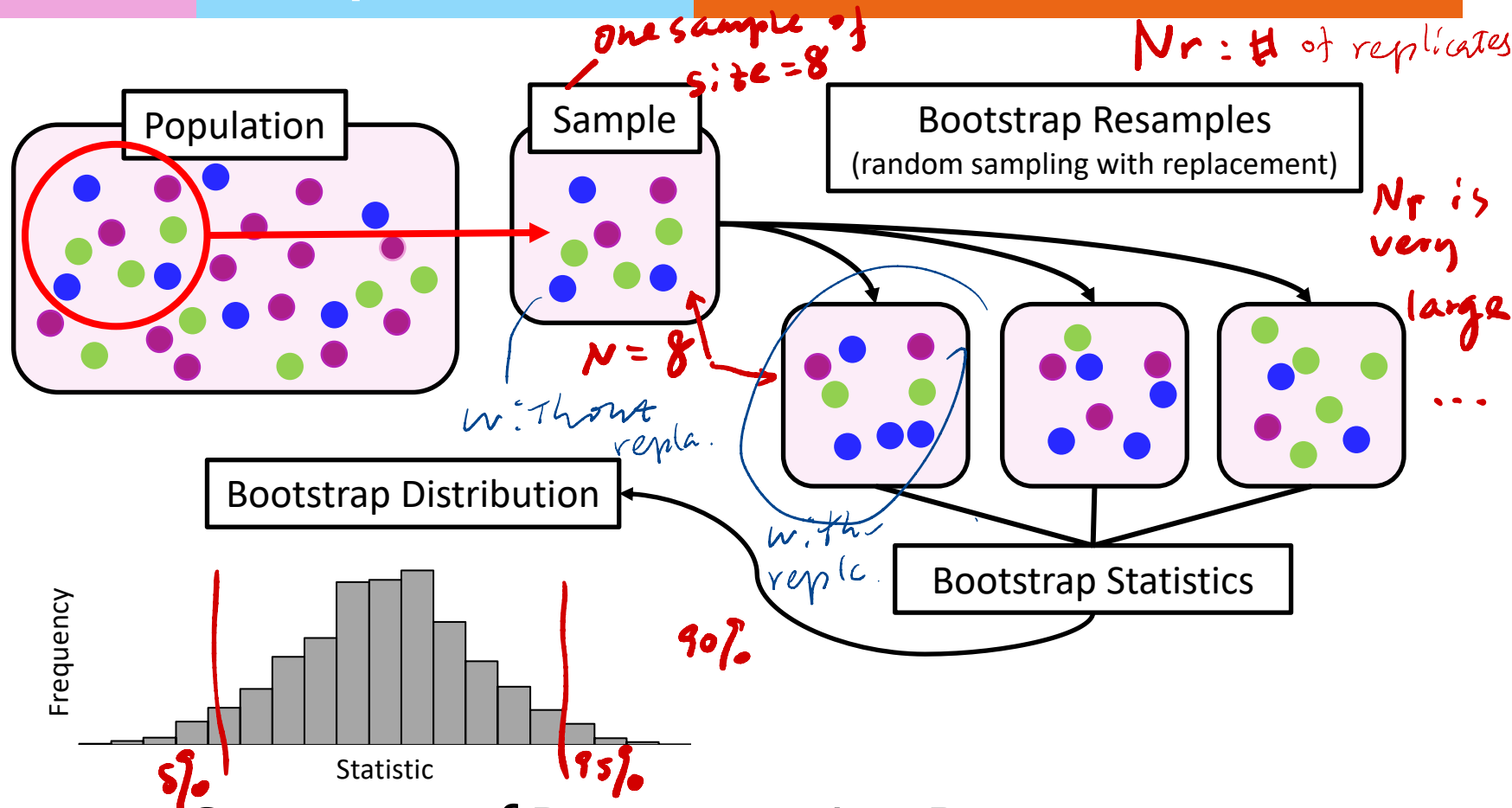
$$C_U = \text{Quantile}(97.5\%)$$

95%

Bootstrap for confidence interval of other sample statistics

- ✱ Bootstrap is a method to construct confidence interval for *any*^{*} sample **statistics** using resampling of the sample data set
- ✱ Bootstrapping is essentially uniform random sampling with replacement on the sample of size **N**

Bootstrap for confidence interval of other sample statistics



Summary of Bootstrapping Process

Example of Bootstrap for confidence interval of sample median

- ✱ The realized sample of student attendance {12,10,9,8,10,11,12,7,5,10}, $N=10$, median=10
↑
- ✱ Generate a random index uniformly from $[1,10]$ that correspond to the 10 numbers in the sample, ie. if index=6, the bootstrap sample's number will be 11.
- ✱ Repeat the process 10 times to get one bootstrap sample

Bootstrap replicate	Sample median
{11, 11, 12, 10, 10, 10, 12, 10, 7, 10}	10

Example of Bootstrap for confidence interval of sample median

- ✱ The realized sample of student attendance $\{12, 10, 9, 8, 10, 11, 12, 7, 5, 10\}$, $N=10$, median=10

Bootstrap replicate	Sample median
$\{11, 11, 12, 10, 10, 10, 12, 10, 7, 10\}$	10
$\{7, 10, 10, 10, 9, 7, 9, 10, 12, 10\}$	10
$\{9, 7, 10, 8, 5, 10, 7, 10, 12, 8\}$	8.5
...	...

Q. How many possible bootstrap replicates?

✱ A. 10^{10} B. $10!$ C. e^{10} D. Others

Bootstrap replicate	Sample median
{11, 11, 12, 10, 10, 10, 12, 10, 7, 10}	10
{7, 10, 10, 10, 9, 7, 9, 10, 12, 10}	10
{9, 7, 10, 8, 5, 10, 7, 10, 12, 8}	8.5
...	...

Example of Bootstrap for confidence interval of sample median

- ✱ Do the bootstrapping for $r = 10000$ times, then draw the histogram and also find the stderr of sample median)

Bootstrap replicate	Sample median
{11, 11, 12, 10, 10, 10, 12, 10, 7, 10}	10
{7, 10, 10, 10, 9, 7, 9, 10, 12, 10}	10
{9, 7, 10, 8, 5, 10, 7, 10, 12, 8}	8.5
...	...

Example of Bootstrap for confidence interval of sample median

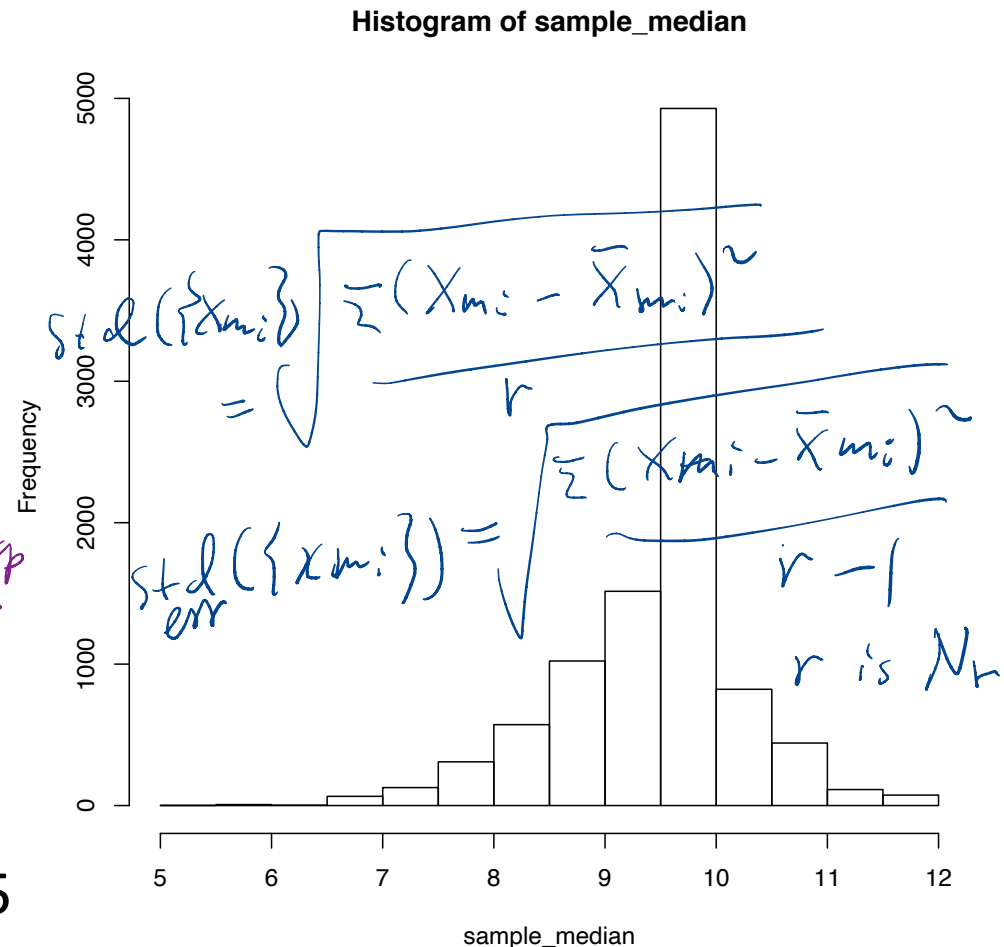
✱ Bootstrapping for $r = 10000$ times, then draw the histogram and also find the stderr of sample median. i : index for bootstrap samples

$$stderr(\{S\}) = \sqrt{\frac{\sum_i [S(\{x\}_i) - \bar{S}]^2}{r - 1}}$$

↳ sample statistic

mean(Sample Median) = 9.73625

stderr(Sample Median) = 0.7724446



Errors in Bootstrapping

- ✱ The distribution simulated from bootstrapping is called empirical distribution. It is not the true population distribution. **There is a statistical error.**
- ✱ The number of bootstrapping replicates may not be enough. **There is a numerical error.**
- ✱ When the statistic is not a well behaving one, such as maximum or minimum of a data set, the bootstrap method may fail to simulate the true distribution.

CEO salary example with larger $N = 59$

✱ The realized sample of CEO salary $N=59$, median=350 K

✱ $r = 10000$

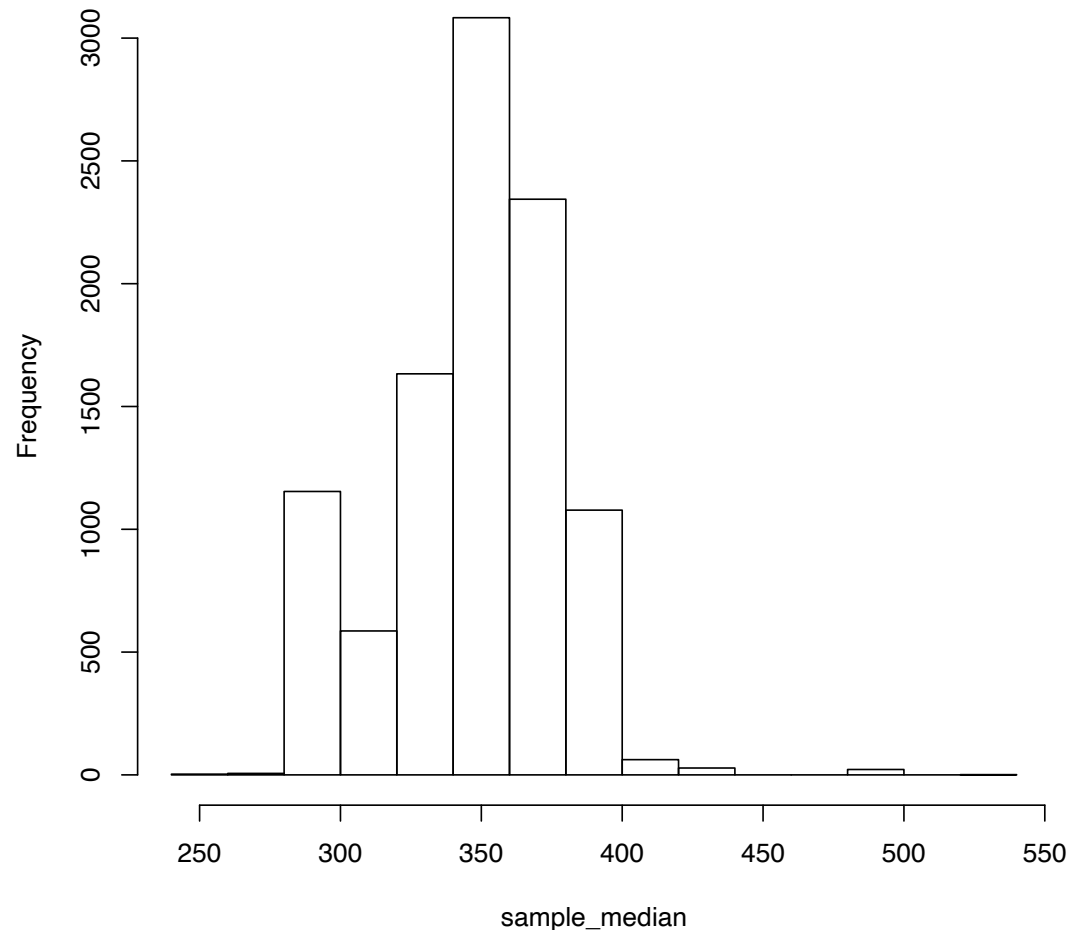
$\text{mean}(\text{Sample Median}) =$

348.0378

$\text{stderr}(\text{Sample Median}) =$

27.30539

Histogram of the Bootstrap sample medians



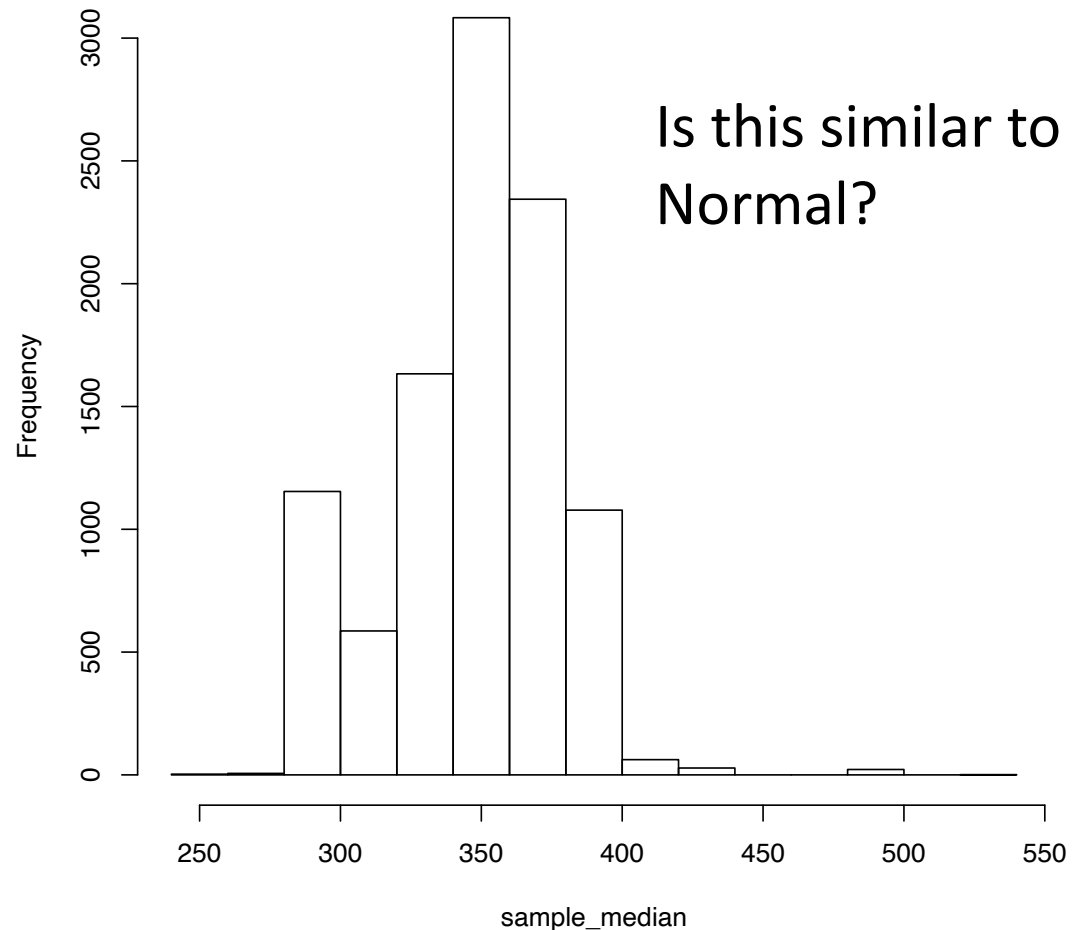
CEO salary example with larger $N = 59$

✱ The realized sample of CEO salary $N=59$, median=350 K

✱ $r = 10000$

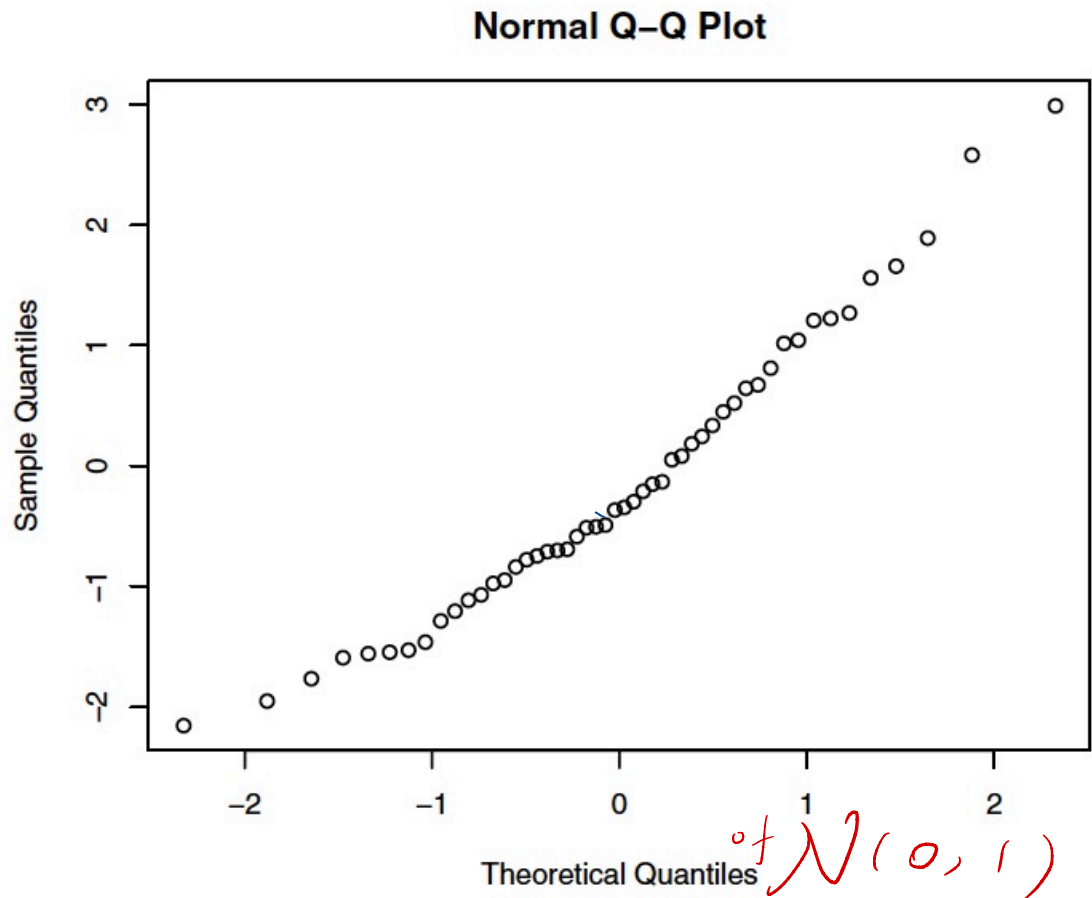
mean(Sample Median) = **348.0378**
stderr(Sample Median) = **27.30539**

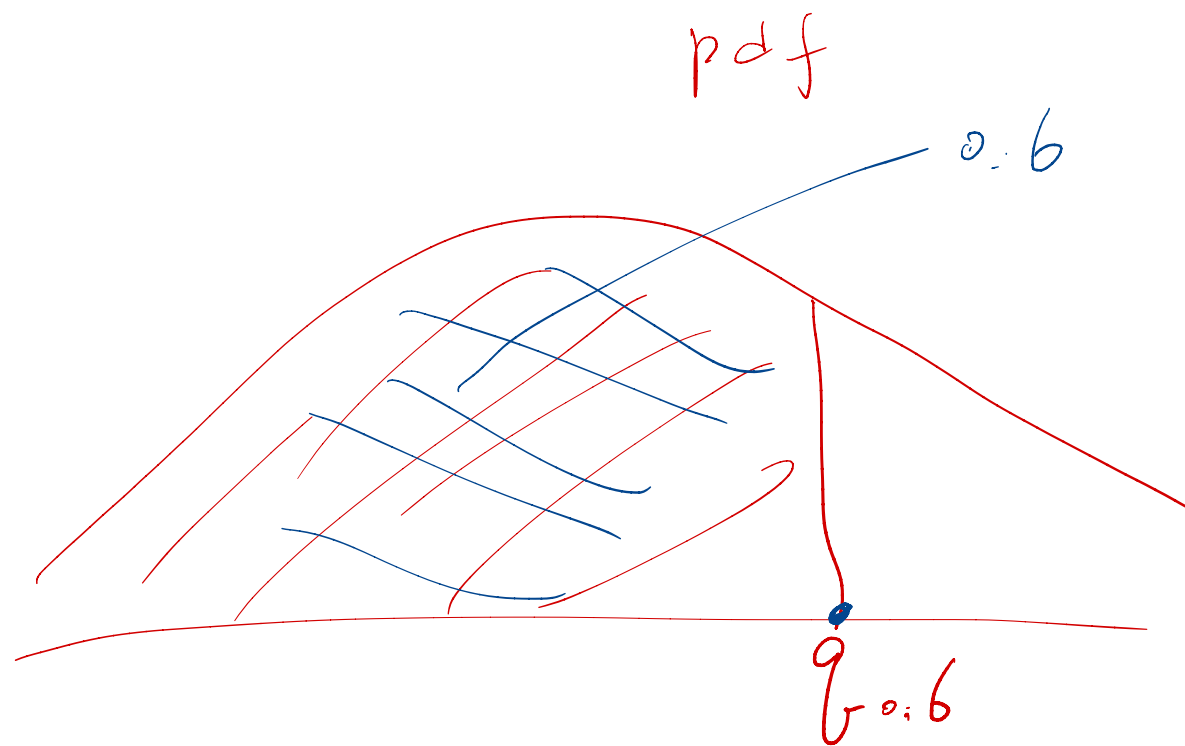
Histogram of the Bootstrap sample medians



Checking whether it's normal by Normal Q-Q plot

- ✱ Q-Q compares a distribution with normal by matching the k th smallest quantile value pairs and plot as a point in the graph
- ✱ **Linear means similar to normal!**



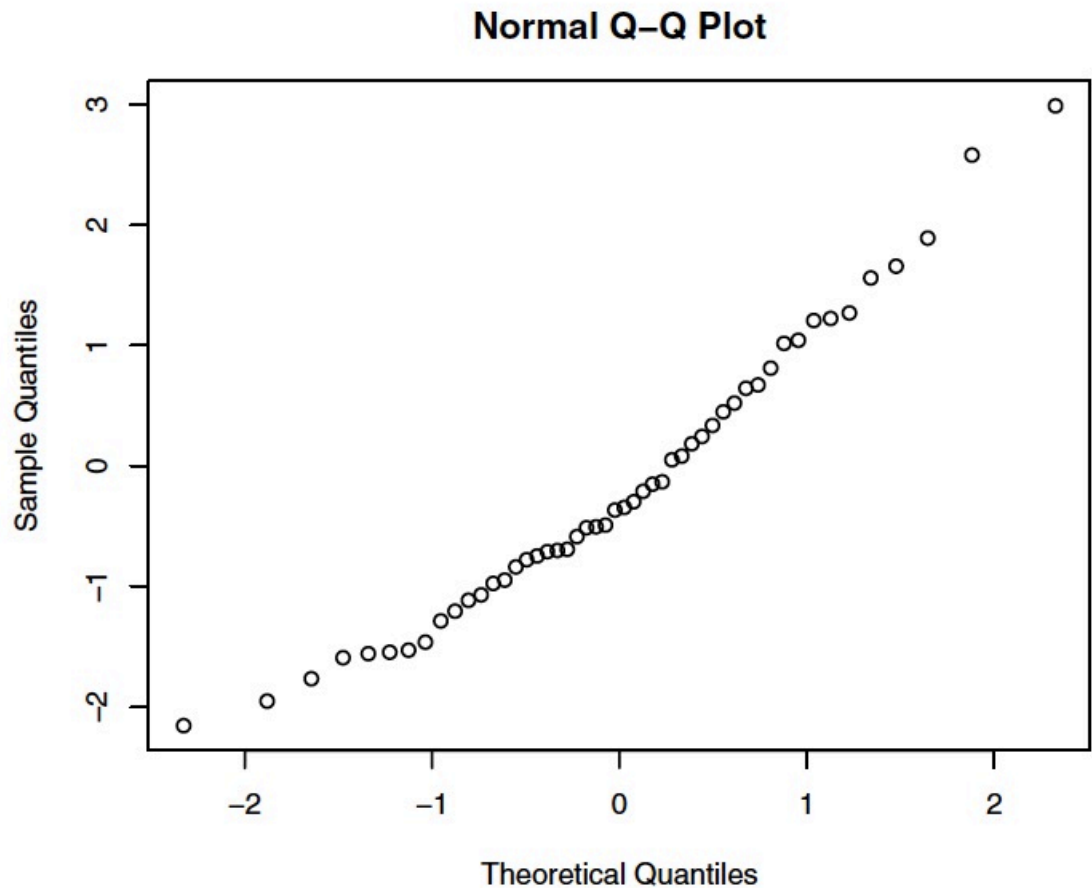


$$P(X \leq q_{0.6}) = 0.6$$

$$CDF(q_{0.6}) = 0.6$$

Checking whether it's normal by Normal Q-Q plot

- ✱ Q-Q compares a distribution with normal by matching the k th smallest quantile value pairs and plot as a point in the graph
- ✱ **Linear means similar to normal!**

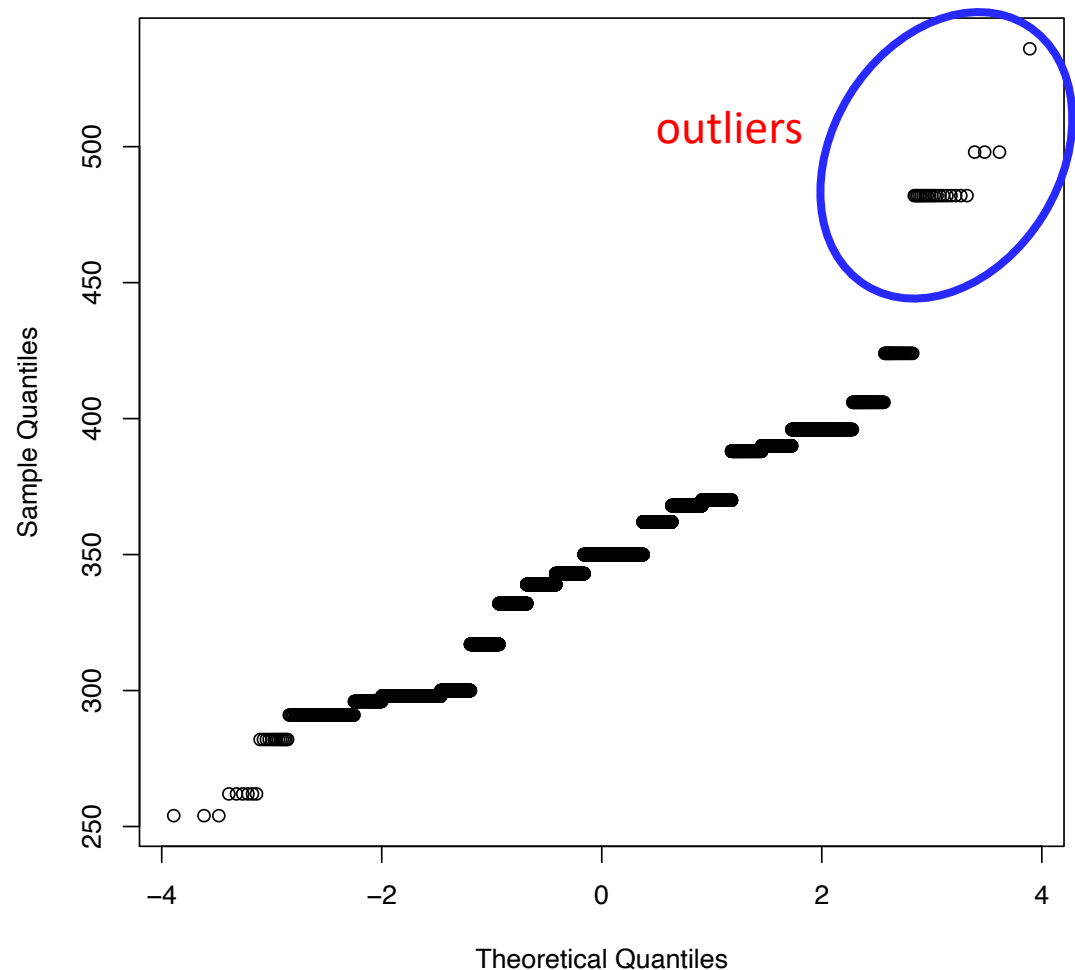


Normal Distribution's Quantile

CEO salary sample median's Q-Q plot

- ✱ Q-Q plot of CEO salary's bootstrap sample medians
- ✱ It's roughly linear so it's close to normal.
- ✱ We can use the normal distribution to construct the confidence intervals

CEO Bootstrap Sample Median Q-Q Plot

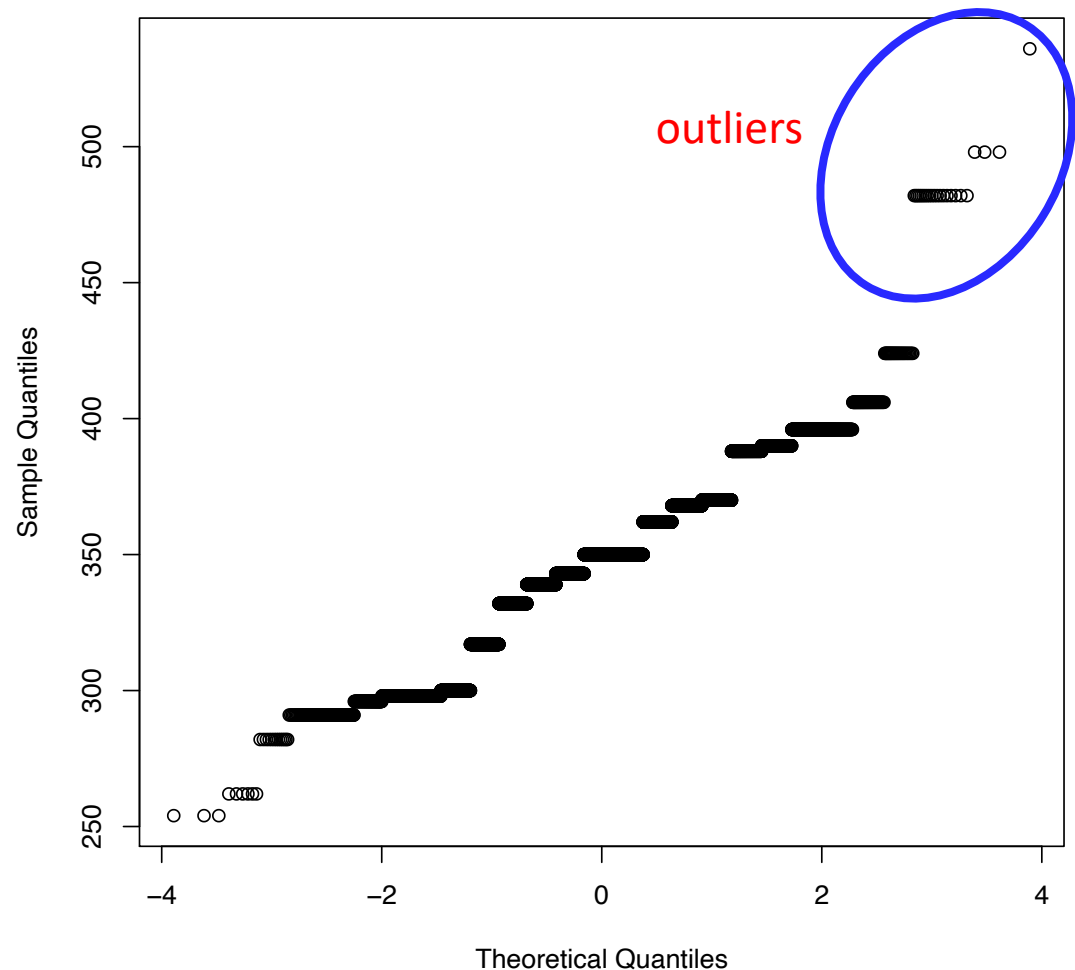


CEO salary sample median's Q-Q plot

✱ 95% confidence interval for the median CEO salary from the bootstrap simulation

✱ $348.0378 \pm 2 \times 27.30539$
 $= [293.427, 402.6486]$

CEO Bootstrap Sample Median Q-Q Plot



Assignments

- ✱ Read Chapter 7 of the textbook
- ✱ Week 8 module on Canvas
- ✱ Next time: more hypothesis testing and MLE

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

See you next time

*See
you!*

